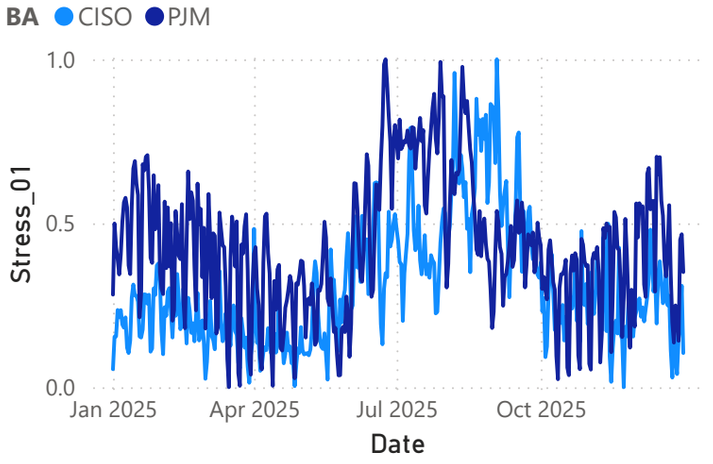
# Electric Grid Stress Analysis - Overview

This dashboard visualizes a machine learning framework built on public grid data classifying daily stress regimes across multiple Balancing Authorities (BAs) using z-scores

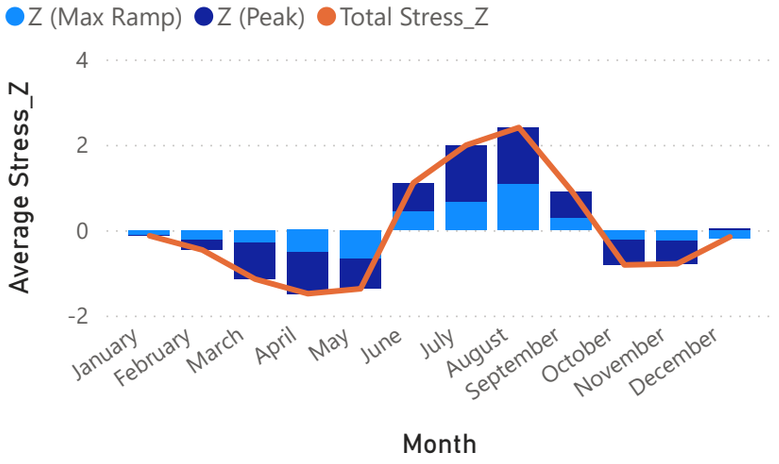
Analytic Structure

Peak = Daily max power demand
Max_Ramp: Daily max power ramp between consecutive hours
Stress_Z = Z_Peak + Z_Max_Ramp (mean = 0, value ≈ std dev)
Stress_01 = normalized Stress_Z by BA (min = 0, max = 1)

## Daily Stress_01




## Average Stress by Month




Note: Stress by Month only informative by separated BAs

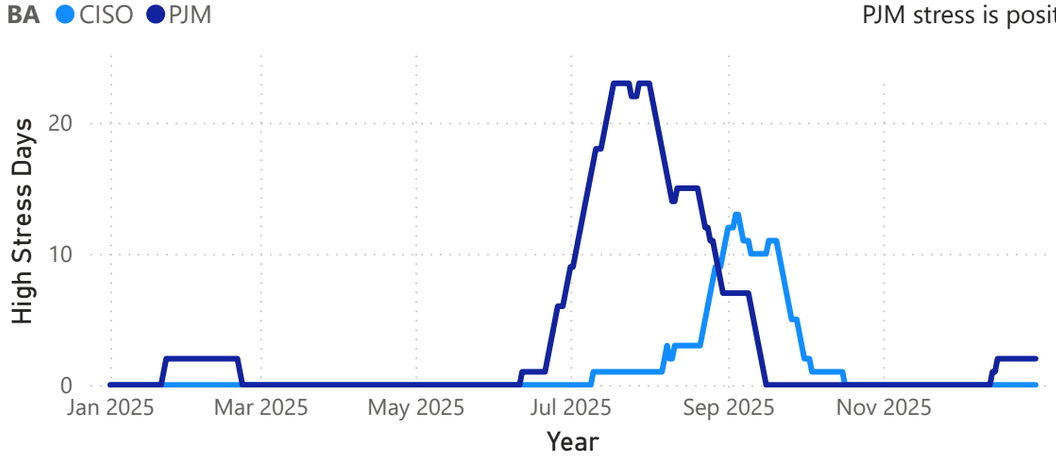
PJM stress is positive in winter months




## Most Stressful Days Table

| BA | Stress_01 | Date |
|---|---|---|
| PJM | 0.98 | June 23 |
| PJM | 1.00 | June 24 |
| PJM | 0.92 | June 25 |
| PJM | 0.89 | July 25 |

## Number of High Stress Days in the Previous 30 Days




Both systems peak in summer, but CAISO's stress concentrates into a tight Aug–Sep window while PJM builds earlier and holds longer

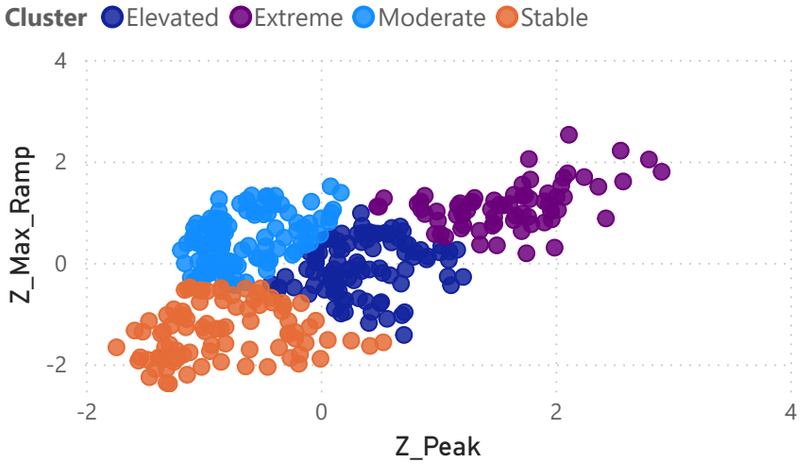
| Stress_Z (Max) | Stress_01 (Avg) | Most Stressed Day | High Stress Days |
|---|---|---|---|
| CISO | CISO | CISO | CISO |
| | | | |
| PJM | PJM | PJM | PJM |
| | | | |

CISO has higher stress days but PJM has more high stressed days

- High Stress Days Threshold: Z_01 > 0.7

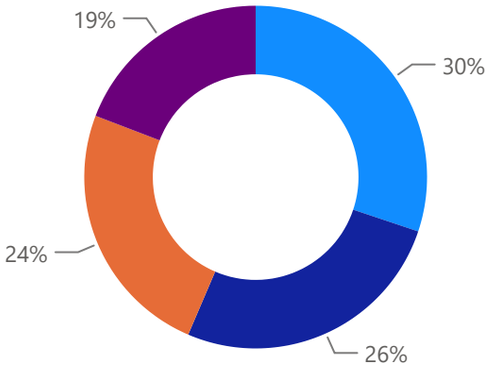
# Electric Grid Stress Analysis - ML Clustering

## Z-Score Clustering




Both BAs cluster similarly with instability increasing along the diagonal.
CISO reaches higher extremes with more spread out datapoints

## Frequency by Cluster

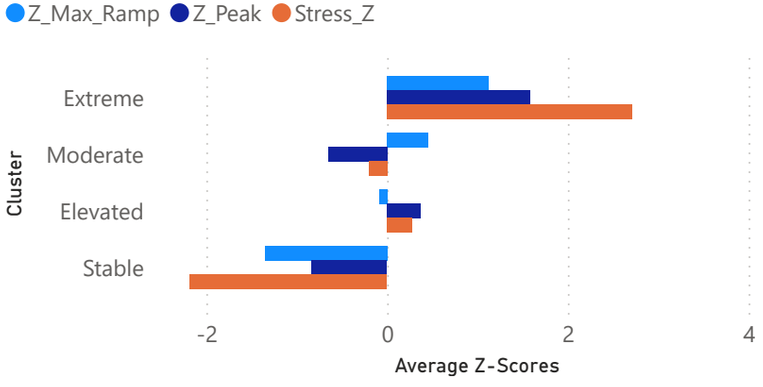






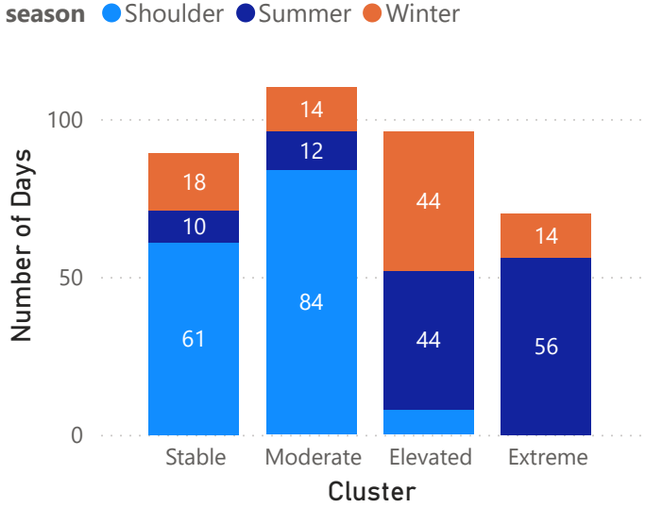
PJM contains more than double the extreme days
CISO contains nearly double the stable days

## Average Z-Scores by Cluster

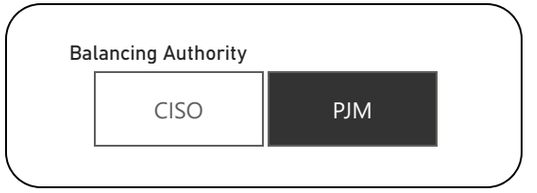



CISO's Extreme cluster combines peak and ramp stress nearly equally
- both must be elevated simultaneously. PJM's Extreme cluster is more peak-led (z_peak 1.58 vs z_ramp 1.12).

## Number of Days by Cluster and Season




PJM has more extreme days (including winter days)




Operators can use this calculator to determine how stressed their grid would be given specified conditions

## Daily Stress Calculator

| Peak Input (MW) | Max Ramp Input (MW) |
|---|---|
| 80000 | 5000 |

| | |
|---|---|
| **Stable**<br>Identified Cluster | **11%**<br>Stress Percentile |
| **-2.20**<br>Input Stress Z | **-2.18**<br>Cluster Mean Stress Z |

Approximate 2025 Ranges (MW)
CISO Peak: 23,000 - 45,000 || CISO Ramp: 1,000 - 5,000
PJM Peak: 78,000 - 160,000 || PJM Ramp: 2,500 - 9,100

| k | CISO | PJM | Total |
|---|---|---|---|
| 2 | 0.51 | 0.42 | **0.93** |
| 3 | 0.40 | 0.41 | **0.81** |
| 4 | 0.40 | 0.41 | **0.81** |
| 5 | 0.38 | 0.39 | **0.76** |
| 6 | 0.37 | 0.41 | **0.79** |
| **Total** | **2.06** | **2.04** | **4.09** |

**Model Validation - Silhouette Scores**

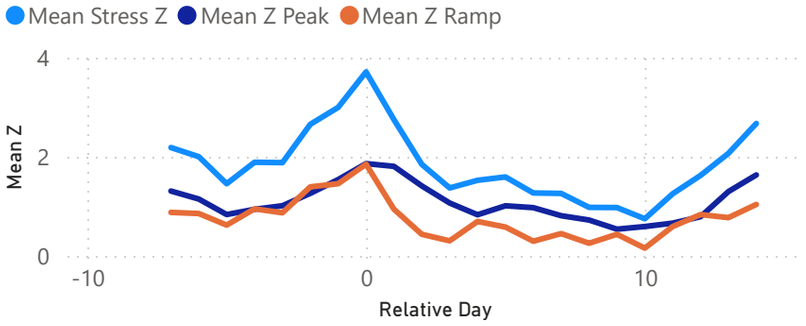
k=4 best for operational interpretability
- four regimes map well to Stable → Extreme
- k=3 collapses the moderate range;

k=6 adds granularity without adding insight

# Electric Grid Stress Analysis - Event Drilldown

**Composite events** - An average of the extreme stress events. How stress builds and resolves around extreme events (day 0 = stress peak)

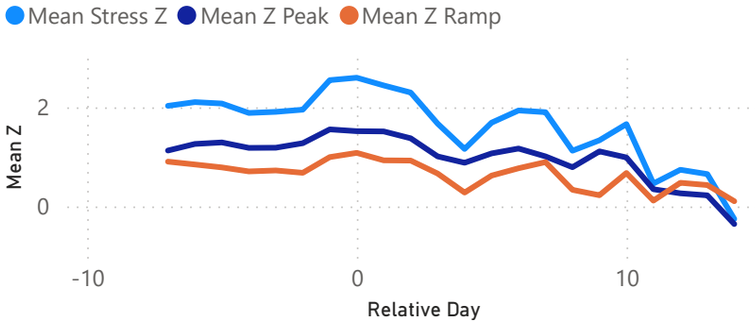
## Composite Extreme Event (CAISO Summer)




Ramp-amplified— solar drop-off drives the evening stress spike

- sharp ramp drives stress on extreme day when solar drop-off occurs

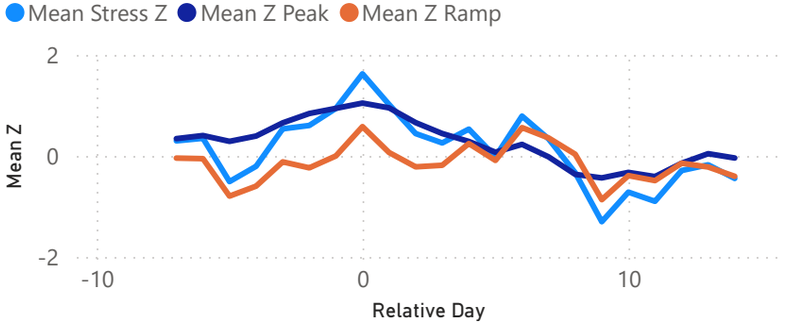
## Composite Extreme Event (PJM Summer)




Load-dominated - slower build-up and drop off due to heat dome effect

- Note CAISO Summer peaks at ~4σ vs PJM at ~2σ

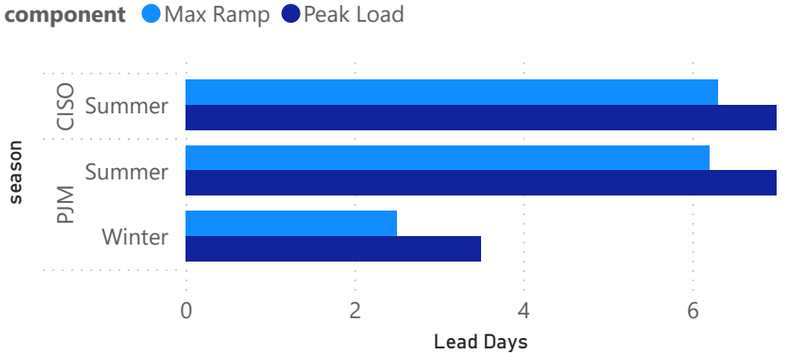
## Composite Extreme Event (PJM Winter)




Load-dominated — quick cold snap raises demand

- ramp signal is weak or negative

## Lead Days to Extreme Event

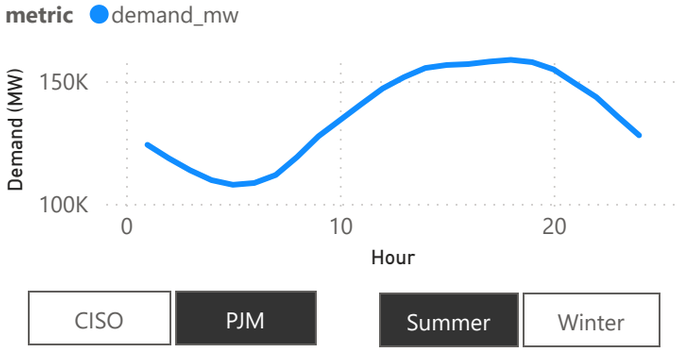



Above: Summer events give much more warning than winter PJM events

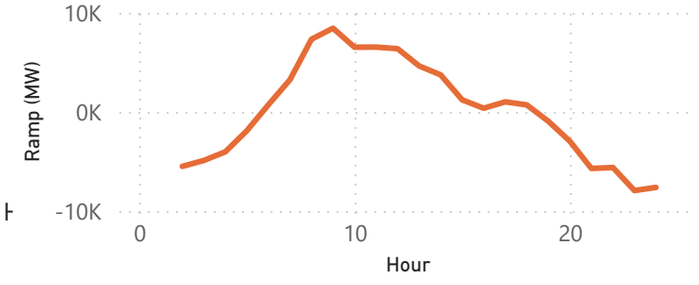
Right - Hourly differences by BA and season demonstrated

- i.e. CISO summer duck curve and PJM winter heating ramps
- CISO solar penetration is ~25% but PJM solar penetration is only ~5%

## Hourly Demand on Extreme Day




## Hourly Ramp on Extreme Day




Summary notes:

- CAISO stress is acute and ramp-amplified — few days, very high peaks. PJM stress is broader and load-led — more days, sustained demand. Different mechanisms require different responses.
- CAISO has a tightly distributed operating regime punctuated by rare violent excursions; PJM has a wider distribution where even extreme events don't deviate as far in normalized terms.
- Compound event — heat wave elevates load for days; solar drop-off then triggers the ramp spike that pushes to Extreme"
- The additive stress formula can produce near-zero stress scores for high peak / low ramp days.
- In practice, such days are rare (r=0.65 between components), but the formula is not designed to flag isolated capacity-only stress.
- All data is based on 2025, but analysis can be applied to other years.